

RUSSIAN TORTOISE • TESTUDO HORSFIELDII

Keep your tortoise the right way for forty years.

A complete, fact-checked printable manual: enclosure size and why a tank is the wrong shape, ultraviolet B (UVB) light you can actually measure, a steppe grazer's real diet, the brumation decision, and the health pages for an animal that will outlast most of your furniture.

INSIDE

- ✓ Enclosure sizing: the minimum floor, the target, and why a tank is the wrong shape
- ✓ A weed list, a grazing plan, and the never-feed list for an arid grazer
- ✓ The brumation decision, in full, with the protocol if you say yes
- ✓ Eight health pages, the legal position, and fillable owner and sitter sheets

41 pages

Intermediate

Print or tablet

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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

If you haven't bought a tortoise yet

Read pages 4 through 13 in order before buying anything; the enclosure and the UVB fixture are the decisions people get wrong most. Then shop from the Budget and Shopping List on page 29. Read page 25 too: the four-inch rule shapes what you can legally buy and from whom, and almost everyone repeats it wrongly.

If your tortoise is already set up

Audit against the Setup Checklist on page 27. Two things are worth checking today whatever else you do: the actual surface temperature of the basking spot with an infrared thermometer, and the age of the UVB bulb, because output falls away months before the light looks dim. Between them they cause most shell deformity and respiratory infection.

Print these first

- | | |
|--|---|
| ☐ Page 27: Setup Checklist & Targets | ☐ Page 34: Pet-Sitter Sheet |
| ☐ Page 28: Emergency & Quick Targets Card | ☐ Page 35: Owner Log |
| ☐ Page 31: Symptom Quick Reference | ☐ Page 36: Equipment, Supplement & Vet Log |
| ☐ Page 32: Daily, Weekly & Seasonal Routine | ☐ Page 13: the never-feed list, for the fridge |

WHY THIS PACKAGE EXISTS

Russian tortoise advice is unusually contradictory, because the animal is genuinely hardy and so a lot of substandard care produces a tortoise that survives for years while slowly deforming. The commonest failures, a glass tank, weak UVB, damp bedding, and too much food, all take years to show. Every figure here is cross-checked against veterinary and specialist tortoise sources, listed on page 39, and every temperature and dimension is given in both imperial and metric.

NOT A SUBSTITUTE FOR

A reptile veterinarian with tortoise experience. This is a husbandry reference for day-to-day care. Anything health-related is a professional call, not a page in this book. Find that vet before you need one, and write the number on page 28. A general small-animal clinic often cannot help with a tortoise, and this species lives long enough that you will need one eventually.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Testudo horsfieldii</i> . Also sold as the Horsfield's tortoise, steppe tortoise, or four-toed tortoise
Native range	Central Asia: Kazakhstan, Uzbekistan, Iran, Afghanistan, Pakistan and western China
Habitat	Semi-arid steppe, rocky desert and dry scrub. A burrowing animal, not a rainforest one
Adult size	5 to 10 inches (13 to 25 centimeters) is the range usually quoted, though the largest verified is about 22 cm (8.7 in) straight carapace length (SCL). Females are usually larger than males
Captive lifespan	40 years or more, with individuals reaching 50 plus. Plan for succession of care, not just your own ownership
Diet type	Herbivore. High fiber, low protein, low sugar, high calcium. Page 11
Indoor enclosure	4 by 2 feet (120×60 cm) is the floor; 7×3.5 ft (210×105 cm), about 24.5 square feet, is the target. Page 5
Enclosure type	Open-topped tortoise table, not a glass tank. This is a ventilation and behavior issue, not a preference
Basking surface	95 to 100°F (35 to 38°C), measured on the surface with an infrared gun
Cool end	70s°F (about 21 to 26°C). Tolerates night lows to 60°F (16°C) without supplemental heat
UVB	Linear T5 HO tube (T5: a fluorescent tube five-eighths of an inch across; HO: high output), target ultraviolet index (UVI) 3.0 to 4.0 at the basking site. Page 7
Humidity	Low ambient, about 30 to 50%, with a moist hide. Page 8
Brumation	Physiologically programmed, 10 to 14 weeks at most, and optional indoors. Page 23
Social	House alone, or with care two females. Never two males. Page 10
Legal status	Colorado bans it; five states require a permit. Page 25. Vulnerable on the International Union for Conservation of Nature (IUCN) Red List, and listed under CITES, the Convention on International Trade in Endangered Species

Cost overview

Initial setup

\$372 to \$775 for a full indoor setup, of which \$125 to \$250 is the table alone. The tortoise itself is \$50 to \$400 depending on age and origin. Page 29.

Ongoing

\$365 to \$715 a year: food, supplements, substrate, UVB replacement, an annual vet check, and electricity, which rises in winter in a cold climate.

THE NUMBER THAT SHOULD DECIDE THIS

Forty years. A tortoise bought for a ten-year-old is likely to outlive that child's own children's interest in it. Rehoming a Russian tortoise is genuinely common and genuinely difficult, so the honest question before buying is not whether you can care for one now but who takes it in 2050. Write that person's name on page 28.

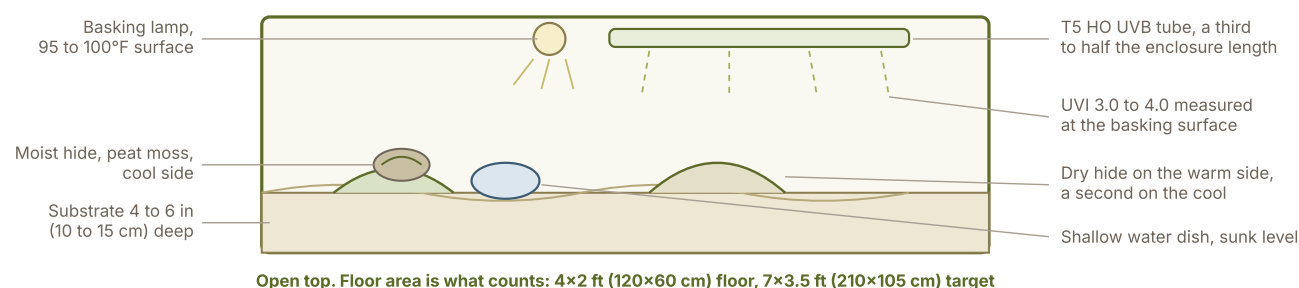
Enclosure size, type & the table question

Floor area, open top, and the reason a glass tank fails this species twice over.

Floor area is the number that matters. The indoor minimum for one adult is **4×2 ft (120×60 cm)**, about 8 square feet (sq ft), and the target is **7×3.5 ft (210×105 cm)**, about **24.5 sq ft**, with about 10 sq ft (1 square meter) more for each additional tortoise. Treat 4×2 ft as the floor below which an adult is not adequately housed, and 7×3.5 ft as the target that a Russian tortoise will actually use, because this is one of the most active tortoises kept and it walks the whole of whatever it is given.

NOT A GLASS TANK

An open-topped tortoise table is strongly preferred, for two separate reasons. **Ventilation:** a closed glass vivarium traps humidity against an arid-adapted animal, which is a direct cause of respiratory infection (page 19). **Behavior:** a tortoise sees through glass but cannot reach what is beyond it, and the result is restless pacing along the wall for hours a day. Neither problem is fixed by buying a bigger tank.



Building or buying

A tortoise table is a shallow open box with solid sides, usually wood sealed against moisture, and it is easy to build and cheap to enlarge. Length matters more than width, and both matter more than height. Do not add a second level: this is a walking animal, not a climbing one, though it climbs better than it looks, which is the subject of page 10.

- **Walls 12 in (30 cm) or more**, solid rather than mesh, with nothing next to them a tortoise can use as a step. Corners are ladders; round them or block them.
- **A sealed, waterproof floor.** Damp substrate against bare wood rots it and grows mold. Pond liner or sealed plywood, with a lip so substrate cannot spill.
- **A genuinely cool end.** If the lamp warms the whole box, the box is too small, whatever the tape measure says.
- **Room for the lamps above it**, at the mounting distances on page 7, without a shelf or a lid in the light path.

HATCHLINGS DO NOT NEED A SMALLER ENCLOSURE

A common piece of advice is to house a young tortoise small so it can find its food and water. It is not necessary and it wastes money on a second enclosure. Give a hatchling the adult footprint with more cover: extra hides, more plants, and food and water in the same place every day. A young tortoise with places to hide uses a large space happily.

Temperature, basking & night lows

A genuine gradient, measured on the surface, in an unusually cold-hardy species.

A tortoise cannot make its own heat, so the enclosure has to offer a range and let the animal choose. What matters is a real gradient across the length of the table: a hot basking spot at one end and a genuinely cool end at the other. If the whole enclosure reads the same temperature, the animal has no way to regulate itself, and that is true whether the reading is too high or exactly right.

ZONE	TARGET
Basking surface	95 to 100°F (35 to 38°C), measured <i>on the surface</i> with an infrared temperature gun, not in the air
Warm end, ambient	Around 80 to 85°F (27 to 29°C)
Cool end	The 70s°F, roughly 21 to 26°C. This end has to actually be cool
Night	A drop to about 60°F (16°C) is fine and is normal for the species. A room that will not hold 60°F at night needs a ceramic heat emitter on a thermostat
Act below	Sustained below 60°F (16°C) outside an intentional brumation, which is where respiratory infection starts. Page 33
Brumation, if chosen	35 to 50°F (2 to 10°C), ideal 41°F (5°C). Page 23

THIS IS ONE OF THE MOST COLD-TOLERANT TORTOISES KEPT

In the wild, Russian tortoises brumate in abandoned rodent burrows 20 to 50 cm (8 to 20 in) down, where the surrounding soil sits at 6 to 10°C (43 to 50°F) and the animal's own body temperature runs 5 to 11°C (41 to 52°F). Overnight temperatures in the +4 to +5°C range (39 to 41°F) are tolerated without harm. That hardiness is real, and it is exactly why owners get away with cold, damp setups for years before the animal shows what it has cost.

Getting the heat right

- **A halogen flood bulb is the right heat source** for a basking spot. It puts out the short-wavelength infrared that penetrates and warms an animal properly, the way sunlight does. A ceramic emitter warms air rather than the animal and is a night or ambient tool, not a basking one.
- **Raise or lower the lamp, do not chase wattage.** Distance is the adjustment. Keep every lamp at least 4 to 6 in (10 to 15 cm) from the highest point the tortoise can reach. A dimming thermostat or a lamp dimmer on the basking bulb lets you hold a temperature through changing room conditions instead of guessing.
- **Never a heat mat under a tortoise.** A tortoise's plastron is not built to take belly heat, it does not register overheating from below, and burns are a real outcome.
- **Two thermometers, permanently.** A digital probe at the cool end and an infrared gun for the basking surface. Stick-on dial gauges are decorative.
- **The basking spot and the UVB tube go at the same end**, so the animal gets heat and ultraviolet (UV) light together, which is how it works outdoors.

HEAT IS THE EMERGENCY IN SUMMER, NOT WINTER

A cold-adapted steppe animal in a sunroom or a sunny room in July is in far more danger than the same animal in an unheated room in January. Outdoors, a pen with no permanent shade is a lethal enclosure. Indoors, check the enclosure's afternoon temperature rather than its morning one, at the hottest point of the year.

UVB, lighting & the equipment that controls it

The one piece of kit you cannot substitute, and the one that fails invisibly.

UVB drives vitamin D3 synthesis, which drives calcium absorption, which builds and maintains a shell. Without it a tortoise develops metabolic bone disease however much calcium is in the food (page 18). What makes it dangerous is that output falls away months before the bulb looks dim, so a fixture that is visibly working can be delivering nothing useful.

REQUIREMENT	DETAIL
Bulb type	A linear T5 HO tube in a reflective fixture: a 10 to 12% desert-strength tube
Length	Roughly a third to half the enclosure length, mounted over the warm end alongside the basking lamp
Target	A UV index of 3.0 to 4.0 at the basking area, measured with a UV index meter
Mounting distance	With mesh in the path, the tortoise's back sits 14 to 16 in (35 to 40 cm) below the tube. With no mesh, 17 to 20 in (43 to 50 cm)
Replacement	Every 12 months at the latest, sooner if the maker rates it shorter, on the calendar rather than when it stops lighting. Write the date on the bulb and on page 36
Visible light	A bright 6,500 kelvin (6500K) daylight lamp alongside. Tortoises are diurnal and a dim enclosure suppresses activity and appetite
Photoperiod	12 hours on, 12 off, on a timer. Lengthen toward 14 in summer and shorten toward 10 in winter to track the seasons

THREE WAYS TO DELIVER NO UVB WHILE BELIEVING YOU ARE

A coil or compact bulb. They concentrate output into a narrow cone, have been associated with eye and skin irritation, and are not adequate for a species that needs a UVI of 3 to 4 across a basking area.

Glass or plastic in the path. Glass removes most of it, which is also why a sunny window is not sunlight to a tortoise. Mesh costs a substantial fraction, which is what the two mounting distances above are for.

A bulb over a year old. Output decays continuously from the day it is switched on.

The equipment that controls the rest

- **A timer** for the lights, so the photoperiod holds whether or not you are home. The cheapest item on the list.
- **A dimming thermostat or lamp dimmer** on the basking bulb, holding the surface temperature through changing room conditions. Mount the thermostat probe at the basking spot, where its maker says, held with a probe clip or suction cup, never tape. An on/off thermostat is the wrong controller for a basking lamp, because it flickers the light.
- **An infrared temperature gun** for the basking surface. Under \$25, and it settles what the other instruments cannot.
- **A digital probe thermometer** at the cool end, plus a hygrometer.
- **A UV index meter**, optional and expensive, and the only way to know rather than assume. Borrow one if a local keeper has it.

SUNLIGHT BEATS EVERY BULB

Unfiltered natural sunlight delivers UVB no lamp matches, so a few supervised hours outdoors in warm weather is worth more than any equipment upgrade. Page 9 covers doing it safely. It does not replace an indoor fixture: the weather does not cooperate for most of the year.

Humidity, substrate & the moist hide

An arid species that still needs somewhere damp, and the two humidity levels it needs.

DRY ON TOP, DAMP UNDERNEATH

Keep ambient humidity at the open surface around **30 to 50%** for an adult, because damp air across the whole enclosure causes respiratory infection; inside the hide and the deeper substrate it runs far higher, up to about 75% at night, as in a burrow. A juvenile is kept moister, about **40 to 75%** ambient, lower by day and higher at night; an adult needs only the humid hide. A tortoise in the wild spends most of its life in a burrow that is far damper than the steppe above it. Build both, and let the animal choose.

In practice: keep the open surface dry and the air moving, keep the deeper substrate slightly damp rather than bone dry, and provide at least one hide packed with moist sphagnum or peat moss on the cool side, remoistened every few days. That combination gives hydration and clean shedding without a permanently damp enclosure, which is what actually causes the respiratory and shell problems on pages 19 and 20.

Substrate

USE	NOTES
50/50 topsoil and coconut coir	The standard mix. Holds a burrow shape, holds a little moisture at depth, and is cheap in bulk. Buy topsoil with no added fertilizer, pesticide or wetting agent
Sandy loam, or a topsoil and play sand blend	Closer to the native substrate. A minority of sand in a soil mix is fine and useful
Depth	At least 4 in (10 cm), which is the minimum for the animal to dig a pallet, and 6 in (15 cm) or more in at least part of the enclosure is better
Avoid: pure sand	Linked to mobility problems and impaction, and it does not hold a burrow
Avoid: heavy wet clay soils	Wrong for a desert species and slow to dry
Avoid: newspaper or rabbit pellets alone	No digging, no thermal buffering, and nothing the animal can do with them
Avoid: cedar and pine shavings, walnut shell, calcium sand	Respiratory irritation, impaction risk, and none of them behave like soil

DIGGING IS NOT DECORATION

Burrowing is how this species buffers temperature and humidity, and a substrate too shallow to dig into removes a behavior the animal actively needs rather than merely enjoys. If the main substrate cannot go deep enough, build a dedicated dig box: a plastic tub of damp soil mix sunk into the enclosure works, and it will be used constantly.

Soaking and water

Keep a shallow water dish available at all times, sunk level with the substrate so the animal can walk in and out and cannot tip. Change it daily; tortoises defecate in water routinely. Beyond that, **soak an adult once a week for 10 to 20 minutes in lukewarm water no deeper than its elbows, and a juvenile 10 to 15 minutes twice a week, never unattended.** Most tortoises drink and pass urates during a soak, which is also when you get the best look at both. Page 24.

Outdoor housing & escape-proofing

The best enclosure this species can have, and the one it will try hardest to leave.

Where climate and security allow, an outdoor pen is the richest environment this species can have: sunlight, real soil to dig, weather, and edible plants growing where it walks. Many keepers treat the pen as standard and the indoor table as the winter arrangement.

Building a pen

ELEMENT	WHAT IT NEEDS
Walls	Solid, not mesh, so it cannot see through and pace. About 2 ft (60 cm) high, with a 4 in (10 cm) lip projecting inward on top; nothing climbable against them
Buried barrier	Walls dug in at least 12 in (30 cm), backed with concrete, paving or mesh, because this species tunnels out of pens that look secure
Water	A dish with a sloped edge it can walk out of, never deeper than the tortoise's own head
Shade	Permanent through the hottest part of the day, from a structure rather than a plant that may be cut back. Heat is the real outdoor danger
Shelter	A dry insulated house it can retreat into in cold or wet weather, with a low entrance and hay or dry leaves
Substrate	Diggable soil, not lawn over hardpan, with a dig area 6 in (15 cm) deep or more
Planting	Edible weeds sown across the pen so it grazes as it walks. Page 12
Cover overhead	A secure top, not just tall walls: dogs, foxes and raccoons come in low, corvids and birds of prey from above. A 2020 study of wild spur-thighed tortoises put 75 millimeters (3 in) of shell as the size above which ravens stopped being a threat, and on open ground ravens took every recorded young-tortoise death. Anything smaller needs a lid and cover to duck under

FIVE WAYS AN OUTDOOR TORTOISE DIES

Heat. No permanent shade at 2 pm in July.

A dog. Shell trauma from a household dog is among the commonest serious injuries in pet tortoises, and it is usually the family's own.

Escape. Digging under, or climbing a corner. Corners are ladders: round them off.

Drowning. A pond, a deep bowl, or an upended tortoise in a dish it cannot right itself out of.

Pesticide, and the wrong plant. Sprayed lawn, treated soil, slug pellets, overspray. Check every ornamental in or over the pen: oleander and the rest of page 13 do not stop being toxic outdoors.

When outdoors is not an option

Indoor keeping is viable for this species for life, which is its main advantage over the larger tortoises. It needs the indoor UVB fixture to be right rather than approximately right, and it makes enclosure size matter more, because no yard is making up the difference. A supervised session outside in warm weather bridges the two.

TAKE THE TEMPERATURE OUT THERE TOO

Pen surfaces reach temperatures a lawn does not. Check a paver, a stone and the shaded soil with the infrared gun on the hottest afternoon of the year. If a surface is too hot for your hand, it is too hot for a plastron.

Handling, temperament & why males live alone

An observation animal with a personality, not a lap animal.

Russian tortoises are active, curious and genuinely responsive to their keepers: many approach confidently, recognize the person who brings food, and rearrange their enclosure overnight. None of that makes them a handling pet. Being picked up is a predator experience for an animal whose entire defense is staying on the ground inside its shell, and over-handling causes real, measurable stress. Keep physical contact purposeful, brief and infrequent, and get the relationship from watching and from feeding instead.

Picking one up, when you have to

- Approach from the front or the side where it can see you, and pause. A hand descending from above is a bird.
- Support the whole shell from underneath with both hands, keeping the animal level and low to the ground.
- Never tip it vertically or onto its back. That compresses the lungs against the organs and is genuinely dangerous, not just unpleasant.
- Never lift by a leg or the front of the shell, and never pull a retracted head or limb.
- Sit on the floor. A dropped tortoise cracks, and a cracked shell is a page 20 problem.
- Wash your hands before and after. Reptiles carry *Salmonella* as normal gut flora, and it is the reason a tortoise is a poor pet for a household with children under five, adults over 65, anyone pregnant or anyone immunocompromised.

SALMONELLA, PLAINLY

Healthy tortoises shed *Salmonella* without being ill, and one negative test clears nothing, because shedding is intermittent. The route is hands, then face or food. Wash after every contact with the animal, its enclosure, dish or substrate. Never clean gear in a kitchen sink, a bathroom sink or a bath people use; keep a plastic tub for it. Do not kiss the tortoise. Children under five should not handle reptiles or anything from their enclosures at all. This is the reason behind the four-inch rule on page 25, and a real risk rather than a regulatory relic.

Housing more than one

House alone by default. This is a solitary species that does not benefit from company the way a social mammal does, and an owner adding a second tortoise for the first one's sake is usually creating a problem. If you do keep more than one:

- **Never two males.** Males are genuinely territorial and will ram, bite legs and flip one another, sometimes fatally. This is not a personality question and it does not resolve with more space.
- **Two females can work**, with caution, in a large enclosure with several hides, several feeding stations and sight breaks between them. Watch for one animal monopolizing the basking spot.
- **A male with a single female is a problem for the female**, who gets pursued and mounted relentlessly. A group of several females to one male spreads that, which is a breeding arrangement rather than a pet one.
- **Never mix species.** Different husbandry needs, and pathogens that one species carries harmlessly can be fatal to another. Page 21.

BETTER THAN THEY LOOK, AT ESCAPING

Despite the reputation, Russian tortoises climb and dig well and are persistent about both. They will scale a corner, push under a barrier, or work at the same weak point for weeks. Assume any enclosure is escapable until it has held for a season, and check the perimeter as a routine job rather than after the first escape.

Diet: what a steppe grazer actually eats

High fiber, low protein, low sugar, high calcium. And less of it than you think.

The dietary rule for *Testudo* species is short: **low protein, low carbohydrate, low sugar, high calcium, high fiber**, and that combination is the primary key to deformity-free growth. In the wild a Russian tortoise eats from a range of over 80 plant species across 27 families, almost entirely weeds, leaves and flowers. It is a grazer of tough, fibrous, low-nutrient plants, and pyramiding and soft shell come from feeding it something richer.

THE QUANTITY NOBODY MENTIONS

In nature this species typically feeds on no more than about **ninety days out of three hundred and sixty five**, and even then spends roughly fifteen minutes grazing a day. Captive tortoises are fed every day, generously, on food far richer than steppe weeds, which is why overfeeding is the single most common husbandry error in this species and why pyramiding is so widespread. Feeding less, and feeding coarser, is closer to correct than most keepers are comfortable with.

What the diet is made of

COMPONENT	SHARE	NOTES
Edible weeds and wild greens	THE CORE	Dandelion leaves and flowers, plantain weed, clover, sow thistle, hawkbit, mallow, chickweed, vetch
Grass hay	ALWAYS AVAILABLE	Timothy, orchard or meadow hay, offered loose. Fiber and a fallback, and a tortoise will graze it steadily
Fibrous leafy greens	SUPPLEMENT	Endive, escarole, radicchio, mustard greens, collard greens, romaine and other dark leaf lettuces, turnip greens, kale, watercress, arugula
Flowers	REGULARLY	Hibiscus, nasturtium, rose petals, pansy, dandelion. Unsprayed
Spinach, chard, beet greens	NEVER STAPLES	High in oxalate, which binds calcium and promotes bladder stones
Commercial tortoise pellets	NOT PRIMARY	Grass-based rehydrated pellets can supplement a winter diet. They should never be the base, and mixed pellets with fruit or grain in them are worse than nothing
Fruit, protein, anything animal	NEVER	Page 13

How to feed

- **Scatter feed rather than using a bowl.** Put greens in several places so the animal has to walk and search, which is closer to grazing and doubles as enrichment.
- **Feed on a flat stone or a tile** where you do scatter feed, so soil and substrate are not swallowed with every mouthful.
- **Adults: feed daily.** There is no published portion size; if flesh crowds the shell openings as it withdraws, feed less. A fasting day once a week does an adult no harm at all and reflects how the animal eats in the wild.
- **Hatchlings and juveniles eat daily and more**, with no fasting days, because they are building shell.
- **Remove uneaten fresh food** the same day. Hay can stay.

Weeds, grazing & growing your own

The cheapest and best food for this species grows in the cracks of a sidewalk.

Supermarket greens are bred to be tender, watery and mild, close to the opposite of what a steppe grazer evolved to eat. Weeds are tougher, higher in fiber and minerals, more varied, and free. A keeper who learns six weeds by sight has better food than one spending \$200 a year on salad, and the variety matters: the wild diet runs to over 80 plant species.

WEED	HOW TO RECOGNIZE IT, AND NOTES
Dandelion	The staple. Deeply toothed leaves in a flat rosette, hollow stem with milky sap, single yellow flower. Leaves and flowers both, in quantity. High in calcium
Plantain (ribwort and broadleaf)	Ribbed parallel-veined leaves in a rosette, with a bare flower spike. Grows in every lawn and path edge. Excellent, and tortoises usually take to it immediately
White and red clover	Three leaflets with a pale crescent, ball-shaped flower. Good, but higher in protein than most weeds, so a component rather than a base
Sow thistle	Tall, hollow stem with milky sap and clasping leaves, small dandelion-like flowers. A favorite. Do not confuse with prickly lettuce, which is also fine
Hawkbit, cat's ear, chickweed	Hawkbit and cat's ear are dandelion lookalikes with hairy leaves and branched stems, and all are fine to feed, which is why the lookalike problem hardly matters. Chickweed is a low sprawling mat with white star flowers: fine, but watery
Mallow	Rounded lobed leaves, pink-purple flowers. Leaves and flowers both, and it is a good hot-weather grower
Vetch, hawksbeard, bramble leaf, hibiscus leaf	All safe and all worth adding for variety

WHERE YOU MUST NOT PICK

Anywhere sprayed with herbicide, pesticide or fertilizer, including a neighbor's overspray. Within a few meters of a road, where exhaust residue and salt accumulate. Anywhere dogs are walked. Any roadside a town, county or contractor maintains. Farmland margins. And never pick something you cannot name: buttercup, ragwort, foxglove, hemlock, ivy, deadly nightshade and daffodil all grow alongside the good weeds. The toxicity data behind lists like these comes from dogs, cats and horses; reptile thresholds are barely studied, which argues for more caution rather than less. Learn buttercup, which grows exactly where you pick clover: glossy waxy flowers where a dandelion is matte, and hand-lobed leaves rather than a flat rosette. If you are not certain, leave it.

Growing your own

A seed tray or two of a tortoise forage mix, sown in rotation, turns cut greens into living plant the animal crops itself, which is as close as a captive setup gets to grazing. Three trays on a windowsill on a two-week rotation keeps a single tortoise supplied through most of the year. Outdoors, sow forage mix straight into the pen and let the animal graze as it walks; that is the single biggest quality-of-life change available for this species. A pot of dandelion and plantain in a sunny corner covers the winter.

DRYING WEEDS FOR WINTER

Dandelion, plantain and clover all dry well. Cut on a dry day, spread in a single layer somewhere warm and airy out of direct sun until crisp, then store in a paper bag rather than plastic. Crumbled into hay it turns a dull winter diet into a varied one, and it costs an afternoon in June.

Fruit, protein, supplements & the never-feed list

The short page that prevents most of the health section.

NEVER FEED A RUSSIAN TORTOISE

Any animal protein. Dog or cat food, boiled egg, cheese, meat, insects, mealworms. Excess protein drives pyramiding and burdens the kidneys, and none of it belongs in a steppe herbivore.

Fruit. Leave it out. It is high in sugar and phosphorus and low in calcium, the opposite of what this animal needs.

Bread, pasta, grain, cereal, cat treats, "tortoise" biscuits.

Iceberg lettuce. Almost no nutrition, and linked to vitamin A deficiency in tortoises fed on it. Page 22.

Toxic plants: avocado, rhubarb leaves, buttercup, ragwort, foxglove, daffodil and other bulbs, ivy, hemlock, nightshade, azalea, oleander, yew.

Anything from treated ground. Page 12.

Why protein and fruit matter so much here

A tortoise's shell grows continuously, in layers, at whatever rate the diet allows. Fed richly, it grows fast and unevenly, and the scutes lift into the raised cone shapes called pyramiding, which is permanent. Fed on tough, fibrous, low-nutrient plants, it grows slowly and smoothly, which is what it does in the wild. The commonest owner instinct, that a growing animal should be fed generously and well, is the exact instinct that deforms this one.

Supplements

SUPPLEMENT	SCHEDULE
Calcium without D3	A light dusting on food at most feedings. This is the daily one
Calcium with D3	Only if your vet asks for it. A tortoise under a T5 HO tube makes its own D3
Reptile multivitamin	Once a week or so, herbivore-appropriate, with no added phosphorus
Cuttlebone	Left in the enclosure permanently, for the tortoise to gnaw at will. Also helps keep the beak in shape

Supplement schedules vary by keeper and by how much natural sun the animal gets, so treat this as the starting point and ask your reptile vet to review it at the annual exam.

IF A VET DOES PRESCRIBE D3, COUNT EVERY SOURCE

Vitamin D3 is fat-soluble and accumulates, and oversupplementation causes its own problems. Read the label on every powder and multivitamin, count the D3 sources you are actually giving, and remember that a strong UVB fixture is one of them.

WATER IS A SUPPLEMENT TOO

A shallow dish always available, changed daily, plus a soak once a week for an adult and twice a week for a juvenile. Hydration does as much for smooth shell growth as any powder, and a dehydrated tortoise is the one that pyramids and the one that forms the urate blockages that become emergencies.

Common mistakes & enrichment

Ten things that go wrong most, and what an interesting enclosure looks like.

THE MISTAKE	WHAT TO DO INSTEAD
A glass tank instead of an open table	Ventilation and pacing, both. Page 5
Buying for the size it is, not the size it becomes	Buy the adult enclosure now
A coil UVB bulb, or one over a year old	Linear T5 HO, replaced on the calendar. Page 7
Measuring air temperature instead of the basking surface	Infrared gun, on the surface. Page 6
Misting the whole enclosure to raise humidity	Dry surface, moist hide. Page 8
Substrate too shallow to dig into	4 to 6 in (10 to 15 cm), or a dig box
Feeding supermarket salad and fruit daily	Weeds, hay and fibrous greens. Pages 11 and 12
Overfeeding a growing tortoise "to help it grow"	Fast growth on rich food is one suspected driver of pyramiding. Page 13
Two males in one enclosure	Never. Page 10
No fecal test on a newly acquired animal	Do it in the first month, symptoms or not. Page 21

What enrichment means for a tortoise

The behaviors this species is built for are walking, grazing, digging and thermoregulating, so enrichment is mostly a matter of building an enclosure those can happen in rather than adding objects to it. A Russian tortoise explores its whole territory every day, pushes things over, and rearranges what it can move. An enclosure that looks tidy the next morning is usually one that is too small or too empty.

Terrain and structure

- Flat stones, cork flats, gentle slopes and low platforms, which do get used rather than walked past.
- Large hollow logs and visual breaks, so the space is not one flat open plain.
- Multiple hides facing different directions, one warm, one cool and moist.
- A deep dig area, or a dedicated dig box.

Routine and change

- Scatter feeding in new places daily or every other day.
- Living plants to crop rather than cut greens in a dish.
- Rearrange non-critical decor every one to four weeks, without moving the hides themselves.
- Supervised outdoor time whenever the weather allows. Page 9.

LEAVE THE HIDES WHERE THEY ARE

Rearranging the scenery is enrichment; moving the shelter is stress. A tortoise builds a mental map of where safety is, and shifting the hide means it has to find safety again from scratch. Change the route, not the destination.

TWO THINGS THAT LOOK LIKE ENRICHMENT AND ARE NOT

A mirror or a reflective surface. A male reads a reflection as a rival and will ram it. **A companion added for company.** This is a solitary species, and a second tortoise brings stress, not company.

Sexing, growth & body condition

Straightforward to sex once mature, and worth weighing every month for forty years.

Sexing

Russian tortoises are not reliably sexable as hatchlings or small juveniles; the differences appear as the animal matures, generally somewhere past about 4 in (10 cm) of shell. Once they do appear they are clear.

FEATURE	MALE	FEMALE
Tail	Much longer and thicker, often carried tucked to one side	Short and blunt
Vent position	Further out along the tail, past the shell margin	Close to the base of the tail
Plastron	Concave, dished inward. This is the most useful cue on an adult	Flat, leaving room for eggs to pass
Carapace shape	Generally flatter	Generally higher domed
Size	Usually smaller	Usually larger
Behavior	Ramming, biting, head bobbing, chasing. Page 10	Digging test nests in season, whether or not a male exists

Growth, and why slow is right

There is no growth chart for this species that is worth printing, because a healthy Russian tortoise grows at whatever rate its diet and its season allow, and the wild pattern is slow and interrupted. Fast growth is a warning rather than an achievement: a tortoise that gains rapidly on a rich diet grows the raised, cone-shaped scutes of pyramiding, which never flatten out again. What you are watching for is **smooth** growth, with new keratin appearing as flat pale rings around the edge of each scute and the surface of the shell staying even.

THE THREE THINGS TO CHECK MONTHLY

Weight, on a digital kitchen scale in grams. Write it down. Trends over years are what this record is for, and the reason a tortoise's owner log runs for decades.

Shell surface. Run a finger across the scutes. Smooth is right; a growing ridge at the scute margins is the beginning of pyramiding and it is a signal to reduce food and increase hydration and exercise now, while it still matters.

Shell rigidity. Press very gently at the rear edge of the carapace and along the plastron. A healthy adult shell does not flex. Any give at all in an animal past hatchling size is metabolic bone disease until a vet says otherwise. Page 18.

Body condition

A tortoise in good condition feels heavy for its size when lifted, the skin around the limbs looks full rather than loose, and the animal can withdraw its head and legs fully without the limbs looking like they barely fit. Signs of poor condition are a light, hollow feel, loose wrinkled skin at the limb sockets, sunken eyes, and lethargy. Because so much of the animal is shell, weight is a better measure than appearance, and the jump-to-conclusion figure worth knowing is that **any sudden weight loss in a tortoise is significant**, since these animals lose weight slowly by nature.

Females, eggs & egg binding

A female with no male in the house can still produce eggs, and she still has to pass them.

A mature female develops and lays eggs on her own hormonal schedule, whether or not she has ever met a male. Those eggs are infertile, but the physical process is identical and so is the risk if she cannot complete it. Every owner of a female should know what a gravid female looks like and what to do when the laying does not happen.

Signs a female is gravid

- **Restlessness and pacing**, often out of character, and walking the perimeter repeatedly.
- **Digging test nests**. Repeated shallow excavations, often abandoned, usually in a warm spot.
- **Reduced appetite** as the eggs take up room.
- **A firm, rounded feel** if a vet palpates the soft area in front of the hind legs. Do not attempt this yourself; pressing on eggs can break them.

PROVIDE A NEST SITE BEFORE SHE NEEDS ONE

A female with nowhere acceptable to dig will hold eggs, and holding eggs is how a straightforward event becomes an emergency. Give her a deep area of slightly damp substrate, deep and loose enough for her to dig a hole as long as her own shell and bury the clutch afterward. There is no honest figure in inches here, because it scales with the animal. Put it in a warm and undisturbed corner. Outdoors, a mound of soil in a sunny spot works. She may reject several sites before choosing one, and being watched puts many females off entirely.

EGG BINDING, OR DYSTOCIA

The test is not how long it has been. A healthy gravid female often stops eating for days or weeks and stays **bright, active and alert**, and a chelonian can hold a clutch for up to six months waiting for a nest site she will accept. What separates that from dystocia is the animal, not the calendar: one gone **weak, flat or unresponsive**, straining with no progress, distended and staying so, or producing fluid or blood from the vent is a same-day case, and so is one that does not improve once a proper nest site appears. Left alone this ends in a ruptured oviduct, peritonitis or death. Treatment ranges from correcting the nest site and giving fluids and calcium, through hormonal induction, to surgical removal of the eggs. It is not something to wait out.

What to do with the eggs

An unmated female is the higher-risk case, not the safer one: it is usually fertilized eggs that get laid, so she is the likelier one to retain a clutch, which is why the nest site goes in early. With no male the eggs are infertile and can be discarded once she has covered the nest. Let her finish: a female interrupted mid-clutch may retain the remainder. If a male has had access they may be fertile, and incubating them is a decision to produce more forty-year animals into a market where rehoming is already hard.

CALCIUM IS WHAT THIS COSTS HER

Producing a clutch draws heavily on a female's calcium reserves, which is the practical link between this page and page 18. A laying female needs her calcium supplementation kept up, a cuttlebone permanently available, and her UVB fixture in date. A female laying repeatedly on a marginal setup is the classic route to metabolic bone disease in an adult that was fine for years.

Health red flags & what to tell the vet

A hardy animal that hides decline for years, then declines quickly.

Almost every health problem here traces back to a handful of husbandry basics, and the pattern is worth learning as a pattern: weak UVB and low calcium cause metabolic bone disease; cold, damp or drafty conditions cause respiratory infection; overfeeding and dry conditions during growth are the suspected drivers of pyramiding; an unscreened new animal brings in parasites and viruses.

SAME DAY, NO WAITING

Open-mouth breathing, gasping, or extending the neck to breathe.

Bubbles or mucus at the nose or mouth, or discharge from the eyes.

Wheezing, clicking or any audible breathing.

Straining while vocalizing, which in a tortoise means a blockage. Page 22.

A female straining or unable to lay. Page 16.

Any crack, puncture or bleeding in the shell, particularly after a dog. Page 20.

Prolapse of anything from the vent.

A tortoise that will not right itself when flipped, or cannot support its own weight.

Sudden collapse or unresponsiveness after a hot day.

Within a few days

- Not eating for more than a few days outside brumation
- Any weight loss, which in this species is always significant
- Soft or flexible shell, or a rear shell margin that gives
- Swelling at the eyelids or beside the head, behind the eyes
- Soft, discolored or foul-smelling patches on the shell
- Diarrhea, or droppings that have changed persistently
- Lethargy, or a tortoise that has stopped walking its territory
- Overgrown beak, or difficulty biting through food

Targets that fix most of it

SETTING	TARGET
Basking surface	95 to 100°F (35 to 38°C)
Cool end	The 70s°F (21 to 26°C); night to 60°F (16°C) is fine
UVB	T5 HO tube, UVI 3.0 to 4.0 at the basking site, replaced every 12 months
Humidity	Dry ventilated surface, moist hide available. Page 8
Diet	Weeds and hay, low protein, no fruit. Pages 11 and 13
Soak	Adults weekly, 10 to 20 minutes; juveniles twice a week

WHAT TO HAVE READY WHEN YOU CALL

Current weight and the weight from a month ago. Basking surface and cool-end readings, actual rather than what the equipment is set to. The age of the UVB bulb. What it eats, in proportions. When it last ate, drank and passed droppings and urates. Whether it has been outdoors, and whether another reptile has been in the house. Photograph the enclosure, the shell top and bottom, and the last droppings.

Metabolic bone disease & pyramiding

The two conditions that are written permanently into the shell.

Metabolic bone disease

Metabolic bone disease (MBD) is among the most common conditions in pet tortoises, and it comes from an imbalance of calcium and phosphorus caused by an improper diet, a lack of adequate ultraviolet light, or inadequate protein intake. Without usable D3 the animal cannot absorb calcium regardless of how much is in the food, so it draws calcium out of its own skeleton and shell to keep blood levels workable.

SIGNS

Misshapen shell. Deformed leg bones. Slow growth. A shell that stays flimsy beyond about six months of age. Weak limbs, an inability to lift the body clear of the ground when walking, or a wobbling gait. In severe cases, fractures that occur with no real trauma. Advanced cases can leave permanently misshapen bones or a permanent shell deformity even after treatment.

Prevention is the whole story: a T5 HO UVB tube at UVI 3.0 to 4.0 replaced yearly, calcium without D3 on most meals, a cuttlebone always available, and a high-calcium low-phosphorus plant diet. Treatment is veterinary, involving injectable calcium and often vitamin D alongside correcting the husbandry. Do not treat a suspected case with supplements alone: calcium given without addressing the D3 pathway is not absorbed.

Pyramiding

Pyramiding is abnormal raised growth of the scutes, so the carapace carries individual sections that look like small pyramids instead of lying flat. The cause is not completely understood. The suspected drivers a keeper controls are excess protein, overfeeding, and too little hydration or humidity during growth. It is largely preventable and **irreversible**: once the scutes are raised they stay raised for the animal's life. A tortoise with mild pyramiding that is otherwise well cared for can live a normal lifespan, so it is a permanent record of past husbandry rather than an ongoing illness. Severe pyramiding does restrict movement and lung capacity.

DRIVER	FIX
Too much food, too often	Feed less when flesh crowds the shell openings, with a fasting day weekly. Page 11
Too much protein: pellets, legumes, any animal food	Weeds and hay. Page 13
Chronic mild dehydration	A weekly soak, twice weekly for a juvenile, water always available, a moist hide. Page 8
An enclosure that is dry everywhere, including the burrow	Dry surface, damp depth
No exercise, in an enclosure too small to walk	Page 5 and page 9

READING THE SHELL AS A HISTORY

A shell records the conditions it grew in. Smooth centers with raised outer rings means the animal grew well early and badly later; the reverse means a bad start that has been corrected. Either way what you can change is the next ring, so treat existing pyramiding as information rather than as something to fix, and check the growing margin monthly.

Respiratory infection & herpesvirus

The urgent one on this list, and the one that can take a whole collection.

Respiratory tract infection

Caused by *Mycoplasma*, other bacteria or viruses, often secondary to vitamin A deficiency. The husbandry triggers are chronically low temperature, whole-enclosure humidity, and drafts, which is why ventilation and dry air are health arguments rather than comfort ones.

SIGNS, AND IT DOES NOT RESOLVE ON ITS OWN

Excess mucus in the mouth, bubbles at the mouth, nose or eyes, nasal discharge, lethargy, loss of appetite, wheezing, extending the neck to breathe, open-mouth breathing or gasping. Any of these is a veterinary appointment now, not a wait-and-see. Treatment is antibiotics orally, by injection or as nose drops; severe cases need hospitalization with fluids and assisted feeding.

While you arrange care, raise the warm end toward the top of the normal range, keep the animal hydrated, and remove any damp or draft. Do not raise the humidity, which is the common and counterproductive instinct.

Herpesvirus, and why quarantine is not optional

THE REASON A NEW TORTOISE NEVER JOINS AN EXISTING ONE

Testudo tortoises are susceptible to epidemics of viral disease and should be treated as potential high-risk carriers of herpes-type organisms. An infected animal is highly contagious and can look healthy for long periods. In the one published outbreak, 2018, the source tortoise had already done a year of quarantine and showed nothing for three years beyond an occasional wet nose; when it broke, 75% of the twenty tortoises exposed died. That is one outbreak, not a rate; losses in an unquarantined group can be near total. Signs: nasal and eye discharge, drooling, white or yellow mouth and tongue plaques, difficulty swallowing, regurgitation, lethargy, neurological signs, pneumonia in severe cases. A mild case can look like slight conjunctivitis, which is why symptom-watching is not screening. No cure, and a survivor carries it for life, shedding after stress and hibernation most of all.

What follows is simple and absolute. A new tortoise is quarantined in a separate room, on separate equipment, for months rather than weeks, and screened by a vet before it goes near another tortoise. Page 21 has the protocol. Never buy on impulse from an expo table where animals from many sources have sat together, and never accept a tortoise from a household whose other animals you have not asked about.

Ranavirus, and mycoplasma

Ranavirus also occurs in chelonians and, like herpesvirus, often proves fatal. *Mycoplasma* causes the chronic upper respiratory disease keepers know: a runny nose that comes and goes for years, worsens under stress, and is hard to clear. It has the same carrier problem, found in captive tortoises with no symptoms at all. All three argue for the same two habits: quarantine properly, never mix species or sources.

A RUNNY NOSE IS NEVER JUST A RUNNY NOSE IN A TORTOISE

Owners call it a cold and wait. There is no tortoise cold that resolves by itself: persistent nasal discharge is an infection, chronic mycoplasmal disease, or vitamin A deficiency. All three need a vet and none improve with time.

Shell rot, trauma & abscesses

The shell is living bone with a skin over it, and it gets infected like any other tissue.

Shell infection, or shell rot

Shell infections, also called shell rot and historically SCUD, or septicemic cutaneous ulcerative disease, are caused by bacteria, fungi or parasites entering through trauma: a scrape, a crack, a burn, or a damp patch that has softened the keratin. An overly damp enclosure is the commonest predisposing condition, which is one more reason to keep the surface dry.

SIGNS, AND WHY THIS IS NOT COSMETIC

Soft, discolored or foul-smelling patches. Pitting, flaking or lifting scutes with damp tissue underneath. A discharge or an unpleasant smell. Deep ulcerations can become life-threatening if not treated aggressively, and shell rot that reaches the bloodstream causes septicemia, which is a genuine emergency. Diagnosis involves microscopic analysis and culture to identify the organism, because a bacterial and a fungal infection need different drugs.

Treatment is cleaning and debridement, medication chosen for the organism, and correcting the husbandry that allowed it. Do not treat a suspected shell infection with a household antiseptic and hope: superficial treatment over an established infection seals it in.

Shell trauma

The commonest cause of shell trauma in a pet tortoise is **a dog in the same household**. A dog does not need to be aggressive: chewing a tortoise like a toy produces punctures and crushing injuries, and a shell is bone with a blood and nerve supply, so it hurts and it bleeds. Any damage can lead to life-threatening bone infection, and severe injuries expose internal organs.

- **Never leave a tortoise and a dog in the same space unsupervised**, including a yard, including a dog that has always ignored it.
- Other sources: a fall from a height, a heavy object, being stepped on, a lamp burn, or a heat mat under the animal.
- Any crack, puncture or bleeding is a same-day veterinary case. Keep the animal clean and dry, do not apply glue, epoxy, resin or tape, and do not submerge it.

Abscesses

Abscesses in tortoises present as **hard, tumor-like swellings** rather than the soft pockets a mammal forms, because the pus is thick and dry. The classic sites are the ear and the mouth: a large swelling on the side of the head just behind the eye is an aural abscess, and swellings in the roof of the mouth also occur. They are frequently related to vitamin A deficiency (page 22). Treatment is surgical drainage or removal, flushing with a medicated solution, culture to identify the organism, and antibiotics. They do not resolve without intervention and they do not drain on their own.

THE MONTHLY SHELL CHECK

Once a month, with the animal on your lap over a towel: look at the whole carapace under good light for soft spots, discoloration, lifting scutes and cracks; turn it over briefly and check the plastron the same way, especially where it meets the substrate; check both sides of the head behind the eyes for swelling; and press gently at the shell margins for give. It takes two minutes and it catches all three conditions on this page early, when they are all treatable.

Parasites, the fecal test & quarantine

Common, mostly invisible, and the reason every new tortoise is screened.

What they carry

Roundworms and protozoa are common gastrointestinal parasites in tortoises, especially wild-caught ones, and often cause no clinical signs at all. Severe infections cause diarrhea, weight loss and failure to thrive. Oxyurid (pinworm) burdens correlate with calcium deficiency, metabolic bone disease and kidney disease, mainly in juveniles, one more reason not to treat a parasite load as separate from husbandry.

HEXAMITA PARVA, THE ONE SPECIFIC TO THIS SPECIES

Hexamita parva is a flagellate protozoan, not a worm, attacking the kidneys and urinary tract rather than the gut. Identified frequently in *Testudo horsfieldii*, untreated it causes rapid weight loss, dehydration and renal failure. **The sign to catch is the urine:** abnormally watery or jelly-like, with soft droppings, means the kidneys are already under strain, and by the time the animal looks unwell it is well established. Diagnosis is a fresh urine or fecal sample under a microscope, fresh because the organism is motile. Treatment is metronidazole at a dose only a vet can set. Isolate immediately: it spreads by cysts through a shared enclosure or dish.

The fecal test

Every newly acquired tortoise gets a fecal test, whether or not it has symptoms. Collect a fresh dropping, bag it, refrigerate it, and get it to the vet within 24 hours; a sample from a soak is fine. Repeat annually, and immediately for unexplained weight loss or a change in droppings. Deworming is prescribed against what is found rather than given routinely: blanket dosing selects for resistance and some anthelmintics are hard on a weak animal.

Quarantine, in full

RULE	WHY
A separate room	Not a second enclosure in the same one. Herpesvirus and mycoplasma travel on air, hands and equipment
Six months, minimum	Not three, and not the 60 to 90 days used for other reptiles: six months to a year, and the 2018 outbreak carrier had already done a year
Separate everything	Dishes, hides, thermometers, tools, substrate, and a wash and change of clothes between
Established animals first, the new one last	Every day, without exception
Testing, not just time	An exam at the start and the end, a fecal test, and where you can get it, a PCR test (polymerase chain reaction, a lab test that finds the virus's genetic material) or serology for herpesvirus, the serology repeated some weeks later since antibodies lag. The 2018 outbreak is the argument: a year of quarantine caught nothing because nobody tested
Weigh weekly	A new animal losing weight in quarantine is the earliest signal there is. Page 15

WILD-CAUGHT ANIMALS, AND WHAT "CAPTIVE BRED" OFTEN MEANS

Large numbers of wild-caught Russian tortoises enter the trade labeled captive-bred or ranches, and a substantial proportion sold as captive-bred probably are not. Imported animals are frequently stressed, dehydrated, injured and diseased, and a heavy parasite burden is the norm. Buy from a breeder who can give you a hatch date.

Vitamin A deficiency, cloacoliths & beak overgrowth

Three conditions from three different corners that show up in the same tortoise.

Hypovitaminosis A

Vitamin A deficiency comes from an inappropriate diet, and the two that cause it are one based on iceberg lettuce and one containing meat. The signs are easy to attribute elsewhere.

SIGNS

Lack of appetite, lethargy, swollen eyelids often with a pus-like discharge so the eyes are partly or fully closed, a swollen ear from an aural abscess seen as a bulge behind the eye, respiratory infections secondary to it, and kidney failure in later stages. Treatment is veterinary vitamin A, sometimes by injection, plus a corrected diet. Do not self-dose: it is fat-soluble and an overdose causes skin sloughing.

Prevention is the diet on pages 11 and 12: dark leafy greens, weeds and a little orange-fleshed vegetable carry the carotenoids a tortoise converts to vitamin A. One eating dandelion and plantain will not be deficient.

Cloacoliths and urate blockage

STRAINING WHILE VOCALIZING IS A TRUE EMERGENCY

A cloacolith is an accumulation of urates, the crystalline portion of a reptile's urine, hardened into a stone in the cloaca. A tortoise straining and making noise while it does is a genuine emergency: this is one of very few sounds a tortoise makes and it means obstruction. Treatment is sedation, pain relief, and breaking the stone down and flushing it out with forceps. Never try to remove anything yourself.

The cause is nearly always chronic dehydration, which is why the soaking routine and the water dish on page 8 matter more than they look. Thick, gritty urates are the warning before the blockage. Page 24. **A bladder stone is the same chemistry further on**, and this species is prone: a 2021 survey of 101 tortoises put Russian tortoises at 8.9% of urolithiasis cases, every stone analyzed pure ammonium acid urate. Straining and reduced appetite were commonest, and some showed nothing until the stone was large. X-ray to find, surgery to remove.

Beak and nail overgrowth

Beak and claws grow continuously and wear down on tough food and abrasive ground. On soft supermarket greens and bare soil, both overgrow. An overgrown beak eventually stops the animal biting properly and it loses weight; a badly overgrown one puts the jaw permanently out of alignment.

- **Prevention:** coarse fibrous weeds and hay rather than soft leaves, a permanent cuttlebone, and flat stones or paving to walk over.
- **Other causes exist**, including metabolic bone disease and an old jaw injury, so a beak that keeps regrowing badly is a symptom rather than a grooming job.
- **Trimming is a vet procedure**, done with a rotary tool. Not clippers, which crack the beak.

SHEDDING, WHICH TORTOISES DO DIFFERENTLY

A tortoise does not shed in one piece. Skin flakes from the neck and legs continuously and scutes shed thin translucent layers as the shell grows; both are normal. Skin retained in tight bands around a limb means dehydration; soft damp tissue under a lifting scute is shell rot. Never peel a scute.

Brumation: the decision and the protocol

Programmed into the species, optional indoors, and dangerous done casually.

Russian tortoises are physiologically programmed to brumate. In their native range they are active only about three months a year, spending the rest in burrows 20 to 50 cm (8 to 20 in) deep where the soil sits at 6 to 10°C (43 to 50°F). Captive tortoises keep the drive, and many slow down and dig in as autumn arrives whatever the enclosure is doing.

THE GENUINE CHOICE INDOOR KEEPING GIVES YOU

Full lighting, heat and feeding year-round holds a Russian tortoise active through winter instead, which is a real option many species do not offer. Wild-caught adults may refuse food or show real stress if stopped from brumating, so the choice is easier with a captive-bred one. Both routes are defensible; the wrong one is half-hearted.

Do not brumate a tortoise that is

- Underweight, or has lost weight this year
- Under about four years old, or newly acquired
- Showing any sign of respiratory infection, however mild
- Carrying an untreated parasite load
- Recovering from any illness or injury
- Not vet-checked in the last few months

The pre-brumation vet check is an annual autumn job, with a fecal parasite screen. A tortoise that goes into brumation sick does not come out. An animal that fails the list above is overwintered warm instead, never below 60°F (16°C), and tried again next year.

The protocol, if you say yes

1. **Weigh and record**, and check the shell, eyes, nose and mouth. This is the baseline every later weighing is read against.
2. **Stop feeding one to three weeks before** the cool-down, the longer end for a bigger animal, holding 70 to 80°F (21 to 27°C) through the fast so digestion finishes. Heat comes off about a week before brumation, not with the food. The gut must be empty: food fermenting in a cold tortoise is a route to serious illness. Keep offering water and soaking.
3. **Wind temperature and photoperiod down gradually**, roughly 5°C (9°F) a week, rather than moving the animal straight into the cold.
4. **Hold it at 35 to 50°F (2 to 10°C), aiming for 41°F (5°C)**, in a ventilated box of dry substrate. Below 35°F or above 60°F is unsafe. **Duration is a ceiling, not a target**: 10 weeks at most for a small animal, 14 for a large one, and waking from 6 weeks on is normal. A dedicated fridge is what most keepers now use, since a garage swings with the weather. It needs a probe thermometer rather than the dial, the door opened 30 seconds daily for air, and a bowl of water in the bottom.
5. **Weigh every one to two weeks**. About 1% of body weight a month is **normal**, not a warning: a 1,000 g tortoise losing around 10 g a month is doing what it should. What ends it early is **total** loss past about 7%, the more conservative of the published thresholds, or any waking, urination, discharge or discomfort.
6. **Warm up gradually** over a few days, then straight back to full basking temperature and UVB.
7. **Soak once alert**, at least 30 minutes, up to twice daily. Rehydration comes before food. If it has not urinated or eaten within about a week, see a vet the same day.

The two that kill are in steps 2 and 4: food left in the gut, and a temperature that drifts. A brumation with no thermometer on it is a gamble.

Reading droppings, urates & hydration

A reptile passes two things, and the second one is the more useful.

A tortoise produces **feces**, the brown or dark green solid, and **urates**, a separate white or off-white portion that is the crystalline waste a reptile excretes instead of liquid urine. It also passes clear liquid urine. Each tells you something different, and the urates in particular are the best hydration gauge an owner has. The soak is when most tortoises pass both, which makes it the easiest time to look.

WHAT YOU SEE	NORMAL	WORTH ACTING ON
Feces	Formed, firm, brown to dark green, often with visible plant fiber. Every few days to about weekly as an adult	Watery or loose persistently, which usually means too much fruit or watery greens, and can mean parasites or infection. Very small and hard means dehydration or too little food. Page 21
Urates	Soft, white, toothpaste-like. Passed with or shortly after feces or during a soak	Thick, pasty, gritty or chalky means dehydration, and left uncorrected a urate or kidney problem, which is the chemistry behind gout. Yellow or orange means the same thing. Hard lumps are the start of a urate blockage, and pink or red means blood
Urine	Clear liquid, colorless	Cloudy, dark, or with grit in it
Whole undigested food	Some visible fiber is normal	Recognizable whole pieces passing through means the gut is not working. Vet
Nothing at all	A few days between droppings is normal, longer in cool weather	More than about a week outside brumation, particularly with a reduced appetite or a bloated feel, needs a vet

GRITTY URATES ARE THE SIGN TO CATCH

Thick, sandy, chalky urates mean the animal is running chronically dry. It is the earliest and easiest warning this species gives, it precedes the urate blockage that is a true emergency, and it is fixed with the two cheapest interventions in this book: more soaking and a genuinely moist hide. Look at the soak water every week.

Hydration

A tortoise from an arid environment is very good at conserving water, which means dehydration is well advanced before it is visible. The signs are sunken eyes, loose or wrinkled skin at the limb sockets, thick urates, reduced appetite, and lethargy. The interventions are all routine rather than emergency:

- **A shallow water dish, sunk level with the substrate, changed daily.** Many tortoises drink far more than their keepers expect once the dish is easy to walk into.
- **A soak once a week for an adult, 10 to 20 minutes, and twice a week for a juvenile, 10 to 15 minutes,** in lukewarm water no deeper than the elbows, never unattended. More often for newly acquired animals and any tortoise with thick urates.
- **A moist hide,** remoistened every few days.
- **Water-containing food.** Fresh weeds and greens carry real moisture; dry hay alone does not.

SOAKING IS ALSO A HEALTH CHECK

A tortoise in a soak is level, still, and easy to look at from every angle. Use the soak: check the eyes and nose, look at the shell top and bottom, feel the limb sockets, and inspect what ends up in the water. It is the single most efficient routine in this book.

The law & the four-inch rule

The species is legal almost everywhere, with one ban people never expect. The rule everyone quotes is not what they think it is.

State by state, the Russian tortoise is among the least restricted animals in the United States. **Of 52 jurisdictions (the 50 states, the District of Columbia and New York City), forty-two place no rule on it at all. Colorado is the one outright ban, five states require a permit, and four jurisdictions attach a condition, among them Hawaii, which bans almost everything yet expressly allows this genus.** The rule that actually shapes how these animals are bought and sold is federal, says nothing about the species, and is misquoted more often than any other rule in the pet trade.

COLORADO IS THE ONE STATE THAT SAYS NO

Colorado allows what its lists name and nothing else. Testudinidae is on Appendix A, so a tortoise looks cleared, but the whole reptile block carries a qualifier and the rule fixes it precisely: the tropical and subtropical region means the band between 26.0 degrees north and 26.0 degrees south. The Russian tortoise is a Central Asian steppe animal from Uzbekistan, Kazakhstan, Afghanistan and Iran, ranging well north of that line, and Colorado Parks and Wildlife (CPW), the state wildlife agency, records *Testudo horsfieldii* on its own herpetofauna list with the finding "native range is outside tropical/subtropical." It fails the appendix on geography, which makes it regulated wildlife, and Colorado has no pet license to issue.

WHAT THE FOUR-INCH RULE ACTUALLY SAYS

It is 21 CFR 1240.62 (Title 21 of the Code of Federal Regulations, section 1240.62), a US Food and Drug Administration (FDA) regulation from 1975 written to reduce *Salmonella* transmission to small children. Its operative sentence is that viable turtle eggs and live turtles with a carapace length of less than 4 inches **shall not be sold, held for sale, or offered for any other type of commercial or public distribution.** Sold, held for sale, offered for distribution. **Nothing about owning one,** and nothing about the species: it catches a hatchling Russian tortoise for the same reason it catches a hatchling red-eared slider.

Then comes the part almost nobody quotes. Subsection (c) lists exceptions, and the second covers the sale, holding for sale and distribution of live turtles and viable turtle eggs **not in connection with a business.** A private keeper rehoming a hatchling is not a business; a breeder selling them is. That clause is why hatchlings move hand to hand through private keepers while shops stock only animals past the four-inch line, and it is the real reason a Russian tortoise you buy is usually already a few years old.

The law, state by state

Every jurisdiction that names the species or attaches a rule to it, and what that means in practice.

WHERE	THE RULE	IN PRACTICE
New Jersey	Not on the exempt species list, so the catch-all permit provision applies	A personal possession permit from Fish and Wildlife, the only one written for the keeper rather than for importation
Hawaii	The genus <i>Testudo</i> is on the List of Conditionally Approved Animals	Allowed, subject to Department of Agriculture conditions. Rare for Hawaii, which works from a closed list where not being named is enough to be banned
Montana	Cleared by family, with the federal size floor written into the statute	Legal above four inches of shell
Rhode Island	All exotic turtles allowed without a permit except named exceptions, which do not include this species	Legal, while the state's own native turtles cannot be kept at all
Colorado	Tortoises are cleared only if native to the tropical and subtropical region, fixed at 26 degrees north to south	Banned. A Central Asian steppe animal fails that test, which makes it regulated wildlife with no pet license available
Massachusetts	The general turtle allowance is subject to a categorical bar on IUCN-listed species	A permit is always required, because <i>Testudo horsfieldii</i> is assessed as Vulnerable
New Mexico, Delaware	Not exempt from the state importation permit	A permit before the animal enters the state. New Mexico group II, Delaware a \$25 Individual permit
West Virginia	W. Va. Code § 20-2-13 bars keeping wildlife brought in from out of state without an importation permit	A permit first. Ask the state Division of Natural Resources in writing before buying
New York City	The Health Code bars turtles under four inches of carapace	An adult is lawful, a hatchling is not, and the animal grows out of the rule
Vermont, Minnesota	Cleared with a condition attached	Vermont requires captive-bred origin paperwork; Minnesota still runs turtles under its older license regime
Everywhere else	No state names the species	Legal in the other 40. Your city ordinance is the only thing left to check

CONSERVATION STATUS, AND WHY IT SHOULD MATTER TO A BUYER

Testudo horsfieldii is Vulnerable on the IUCN Red List and CITES-listed, and large numbers of wild-caught animals have entered the trade labeled captive-bred or ranched. A cheap adult at an expo is very often a wild-caught import, with the parasite load and disease risk that carries (page 21). Captive-bred costs more and does not draw on a wild population.

Rules change, and this page is a summary rather than legal advice. Confirm your own state and city before buying, and check import rules before crossing a border with a CITES-listed animal.

Setup checklist & targets

Print this one. Audit an existing setup against it, or buy against it.

Enclosure

- ☐ Open-topped tortoise table, not a glass tank
- ☐ Floor at least 4×2 ft (120×60 cm); 7×3.5 ft (210×105 cm) is the target
- ☐ Walls 12 in (30 cm) or more, nothing climbable against them
- ☐ Substrate 4 to 6 in (10 to 15 cm), topsoil and coir mix, damp at depth
- ☐ Two hides at least: one warm and dry, one cool and moist
- ☐ Shallow water dish sunk level, changed daily
- ☐ Flat stone or tile as a feeding surface
- ☐ Cuttlebone left in permanently

Heat and light

- ☐ Halogen basking lamp, surface reading 95 to 100°F (35 to 38°C)
- ☐ Cool end genuinely in the 70s°F (21 to 26°C)
- ☐ No heat mat; every lamp 4 to 6 in (10 to 15 cm) above its highest reach
- ☐ Dimming thermostat on the basking bulb, probe clipped at the basking spot
- ☐ Linear T5 HO UVB tube, a third to half the enclosure length
- ☐ UVI 3.0 to 4.0 at the basking site; mounting distance checked
- ☐ Bulb replacement date written on the bulb and on page 36
- ☐ Timer set for 12 hours on, 12 off

Instruments, food and care

- ☐ Infrared temperature gun for the basking surface
- ☐ Digital probe thermometer at the cool end, plus a hygrometer
- ☐ Digital kitchen scale reading in grams
- ☐ Calcium without D3 and an herbivore multivitamin; D3 only if the vet asks
- ☐ Weeds or a forage seed mix growing, plus grass hay always available
- ☐ Soaking routine set: adults weekly, juveniles twice a week
- ☐ Reptile vet with tortoise experience, number on page 28
- ☐ Fecal test booked for the first month

Targets at a glance

Basking surface	95 to 100°F (35 to 38°C)
Cool end and night	70s°F (21 to 26°C); night to 60°F (16°C) is fine
UVB	UVI 3.0 to 4.0 at the basking site, bulb replaced every 12 months
Humidity	Dry ventilated surface with a moist hide. Page 8
Diet	Weeds and hay, low protein, no fruit; feed less if flesh crowds the shell openings
Vet	Fecal test on arrival and annually, plus a yearly wellness exam

Emergency & quick targets card

Fill this in today. Cut it out or photograph it and keep it on your phone.

Reptile vet _____

Phone _____ Hours _____

After-hours / exotics emergency clinic _____

Phone _____ Drive time _____

Backup caregiver _____ Phone _____

Long-term caregiver, if I cannot keep it _____

Tortoise name _____ Hatch or purchase date _____

Sex _____ Baseline weight (g) _____

Go now, do not wait

- Open-mouth breathing, gasping, neck extended
- Bubbles or mucus at nose or mouth
- Straining while making noise. Page 22
- A female straining or unable to lay. Page 16
- Any shell crack, puncture or bleeding, especially after a dog
- Prolapse from the vent
- Collapse or unresponsiveness after a hot day
- Cannot right itself, or cannot lift its body to walk

WHILE YOU TRAVEL

A ventilated box lined with damp paper towel, no substrate to shift around, and no water dish to spill. Keep it warm, around 75 to 80°F (24 to 27°C), and dark and quiet. In cold weather use a wrapped hot water bottle beside the box, never touching the animal. Take a photograph of the enclosure and the last droppings, and bring the UVB bulb's age, the basking temperature reading, and the current weight.

Targets, restated

Enclosure floor	4×2 ft (120×60 cm) minimum, 7×3.5 ft (210×105 cm) target, open top
Basking surface	95 to 100°F (35 to 38°C), measured on the surface
Cool end / night	70s°F (21 to 26°C); night to 60°F (16°C). Act if sustained below 60°F
UVB	UVI 3.0 to 4.0; T5 HO tube replaced every 12 months
Substrate	4 to 6 in (10 to 15 cm), damp at depth, dry at the surface
Soak	Adults weekly, 10 to 20 minutes; juveniles twice a week
Brumation	35 to 50°F (2 to 10°C), ideal 41°F, 10 to 14 weeks at most, only if healthy. Page 23

UVB bulb fitted _____ Replace by _____

Last fecal test _____ Last wellness exam _____

Budget & shopping list

Itemized, and the lines add up. US retail, September 2026.

One-time setup

ITEM	LOW	CONSIDERED
Tortoise table, bought or built, 4×2 ft or larger	\$125	\$250
T5 HO UVB fixture and bulb	\$60	\$130
Halogen basking lamp, fixture and a dimming thermostat	\$50	\$105
Timer	\$10	\$20
Infrared temperature gun	\$18	\$30
Digital probe thermometer, hygrometer and substrate fill	\$40	\$90
Hides, cork, flat stones and a dig box	\$30	\$70
Water dish, feeding tile and a digital gram scale	\$24	\$50
Calcium without D3, multivitamin, cuttlebone	\$15	\$30
Equipment subtotal	\$372	\$775
The tortoise, hatchling captive-bred to adult	\$50	\$400
First vet exam plus a fecal test	\$75	\$200
Total to get started	\$497	\$1,375

Ongoing, per year

ITEM	LOW	CONSIDERED
Greens, weeds and grass hay	\$140	\$280
Supplements	\$20	\$35
Substrate replacement	\$40	\$90
UVB bulb, replaced annually	\$45	\$90
Electricity for heat and light	\$60	\$120
Annual vet wellness check with a fecal test	\$60	\$100
Annual total	\$365	\$715
Which is, per month	\$30	\$60

THE FORTY-YEAR NUMBER

At \$365 to \$715 a year, a Russian tortoise costs roughly **\$15,000 to \$30,000 over a forty-year life**, before a single illness. Sick or emergency care with tests and medication commonly runs \$150 to \$300 or more per episode, and this animal will have episodes. Nobody budgets a tortoise this way and everybody should: the purchase price is the least significant number on this page.

Electricity moves most: a cold house costs meaningfully more to keep a basking spot in, and a tortoise outdoors for half the year costs almost nothing to heat. Save on the table, which is a plywood box anyone can build, on substrate bought as topsoil rather than as a reptile product, and on food, which grows free in a lawn (page 12). Do not save on the UVB fixture, the thermostat, or the vet: a coil bulb costs a fraction of a T5 kit and produces the deformity on page 18.

First 30 days checklist

In order. Most of week one is letting it settle and watching what comes out.

Before it arrives

- ☐ Enclosure built, filled, lit and running for several days
- ☐ Basking surface measured with the infrared gun and adjusted to 95 to 100°F
- ☐ Cool end confirmed genuinely cool with a probe thermometer
- ☐ UVB mounted at the right distance, date written on the bulb
- ☐ Reptile vet found and the number written down
- ☐ Quarantine room ready if you already keep reptiles. Page 21

Days 1 to 7

- ☐ Weigh on arrival and write the number down. This is the baseline for the next forty years
- ☐ Handle only to move it in and to soak it. Otherwise leave it alone
- ☐ Soak this week, 10 to 20 minutes in lukewarm water no deeper than the elbows, twice for a juvenile. Newly moved tortoises are usually dehydrated
- ☐ Offer food daily and do not worry if it is refused for the first few days
- ☐ Watch for the first droppings and urates, and keep a fresh sample for the vet
- ☐ Check the basking temperature again with the animal actually in the enclosure

Days 8 to 14

- ☐ Veterinary health check plus a fecal test, symptoms or not. Page 21
- ☐ Ask the vet to confirm sex if the animal is large enough, and to review the supplement schedule
- ☐ Weigh again. A newly moved tortoise often dips slightly, then recovers
- ☐ Switch the diet gradually toward weeds and hay if it arrived on something else. Page 11

Days 15 to 30

- ☐ Soaking routine established, and the weekly weigh-in logged. Page 35
- ☐ Scatter feeding started, in different places each day. Page 14
- ☐ Weeds identified in your own area, or a forage tray sown. Page 12
- ☐ First monthly shell check done. Page 20
- ☐ First supervised outdoor session if the weather allows. Page 9
- ☐ Long-term caregiver named and written on page 28

WHAT NORMAL LOOKS LIKE BY DAY 30

Out and walking the enclosure in the morning, basking for a stretch, eating within a few hours of food appearing, digging into the substrate or a hide by late afternoon, and producing formed droppings with soft white urates every couple of days. A tortoise that hides constantly, refuses food for weeks, or has not passed anything is not settling in, it is unwell. Page 17.

Symptom quick reference

What you see, what it might be, and how fast to move.

WHAT YOU SEE	MOST LIKELY CAUSES	ACT
Open-mouth breathing, gasping, neck extended	Respiratory infection. Page 19	Now
Bubbles or mucus at nose or mouth	Respiratory infection, mycoplasma, vitamin A deficiency	Now
Straining and making noise	Cloacolith or urate blockage. Page 22	Now
Female straining, digging then stopping, off food	Egg binding. Page 16	Now
Shell cracked, punctured or bleeding	Dog, fall, crush. Page 20	Now
Prolapse from the vent	Straining, blockage, egg binding, parasites	Now
Collapsed or unresponsive after a hot day	Overheating. Move to shade, cool gradually, then vet	Now
Cannot lift its body to walk, or wobbles	Metabolic bone disease. Page 18	Same day
Shell soft or flexible past hatchling size	Metabolic bone disease	Same day
Swollen eyelids, eyes closed, pus-like discharge	Vitamin A deficiency. Page 22	Within days
Hard swelling behind the eye	Aural abscess. Page 20	Within days
Soft, discolored or smelly patches on the shell	Shell rot. Page 20	Within days
Persistent runny nose	Never just a cold. Infection, mycoplasma, or vitamin A	Within days
Steady weight loss	Parasites, including <i>Hexamita</i> , dental or beak problems, kidney disease. Page 21	Within days
Thick, gritty, chalky urates	Dehydration heading toward a blockage. Page 24	Soak now, vet if it persists
Watery droppings	Too much fruit or watery greens, parasites, infection	Correct diet, then vet
Nothing passed for over a week outside brumation	Impaction, blockage, dehydration, too cold	Within days
Raised, cone-shaped scutes appearing	Pyramiding. Reduce food, add water and exercise. Page 18	Change husbandry now
Overgrown beak, difficulty biting	Soft diet, MBD, past injury. Page 22	Within weeks
Skin retained in tight bands on a limb	Dehydration. Soak daily and check the moist hide	Within days
Refusing food in autumn, digging in, slowing down	Brumation instinct. Page 23	Decide, do not drift
Pacing a wall for hours	Glass sides, or an enclosure too small. Page 5	Change the enclosure
Ramming or biting another tortoise	Male territoriality. Separate. Page 10	Now

Daily, weekly & seasonal routine

Ten minutes a day, half an hour a week, and one decision each autumn.

Every day

- ☐ Fresh food, scattered in new places, on a tile or stone
- ☐ Water dish emptied, rinsed and refilled
- ☐ Remove yesterday's uneaten greens and any droppings
- ☐ Look at the tortoise properly: gait, eyes, nose, and whether it basked
- ☐ Check the basking surface reading and the cool end
- ☐ Damp the moist hide if it has dried out

Every week

- ☐ Soak (twice for a juvenile), 10 to 20 minutes, and use it as a health check. Page 24
- ☐ Weigh and log: weekly for a juvenile or new arrival, monthly for a settled adult. Page 35
- ☐ Top up hay and refresh the forage trays
- ☐ Turn and refresh the substrate surface; replace any soiled patch
- ☐ Rearrange non-critical decor, leaving the hides where they are
- ☐ Walk the perimeter of any outdoor pen looking for digging

Every month

- ☐ Shell check, top and bottom, under good light. Page 20
- ☐ Check the shell margin for the ridging that starts pyramiding. Page 15
- ☐ Full substrate turn or partial replacement, and a wash of dishes and hides
- ☐ Review the weight log as a trend rather than as this month's number
- ☐ Check the UVB bulb's replacement date. Page 36

Seasonal

WHEN	WHAT CHANGES
Spring	Weeds return. Start picking, sow forage trays, and begin outdoor sessions once daytime temperatures allow. A tortoise coming out of brumation is soaked once alert, at least 30 minutes, up to twice daily
Summer	The dangerous season. Check the enclosure and the pen at the hottest part of the afternoon, confirm permanent shade, and never leave a tortoise in a sunroom, greenhouse, car or unshaded pen. Page 6
Autumn	The brumation decision, made deliberately rather than drifted into. If yes, the fasting period starts now. If no, keep heat, light and feeding fully on and expect the tortoise to slow down anyway. Page 23
Winter	Either the animal is in brumation and weighed every one to two weeks, or it is fully active and the enclosure is fighting a cold house. Watch for drafts, and expect the electricity line on page 29 to rise
Yearly	UVB bulb replaced on the date, not on failure. Vet wellness exam with a fecal test. Substrate fully replaced. Enclosure and pen checked for digging damage and rot

Power outages, travel & transport

A reptile depends on an outlet for the one thing it cannot make: body heat.

A healthy adult tolerates one ordinary cool night without harm. The risk is duration, not a single low reading, and this species has more margin than almost any reptile kept: it **tolerates 60°F (16°C) comfortably**, and the point to act is a sustained spell below 60°F outside brumation. Held below its preferred range for more than a day or two, a reptile becomes more susceptible to bacterial infection and stops eating, which is the setup for the respiratory infection on page 19.

During an outage

- **Move the enclosure to the warmest interior room**, away from windows and exterior doors.
- **Drape thick towels or a blanket over the enclosure, leaving a vent gap**, to hold in the heat that is left.
- **A wrapped hot water bottle or a commercial hand warmer beside the animal**, never touching it directly and never underneath it.
- **Stop feeding** if the enclosure is genuinely cold. A reptile cannot digest below its working temperature, and food sitting undigested in a cold gut is a real problem. This is the opposite of the rule for a mammal or a bird.
- **Keep water available**, and keep soaking on the normal schedule with warm water heated another way.
- **No candles, gas stoves or open flames near the enclosure**. The fire and carbon monoxide risk outweighs the cold, which this species handles well.

Outage kit, packed before you need it: two or three hand warmers or a hot water bottle, two thick towels, a battery or wind-up thermometer, a ventilated tub sized for the animal, and your reptile vet and 24-hour exotics numbers written on paper rather than saved on a phone that may die.

Travel and being away

TRIP LENGTH	WHAT A HEALTHY ADULT NEEDS
1 to 2 nights	Often fine alone. Lights on timers, thermostat running, water dish full, and a generous scatter of greens and hay before you leave
3 to 7 nights	A sitter every 1 to 2 days for fresh food and water, a look at the droppings, and a temperature check
Over a week	A daily or every-other-day sitter who has done a practice visit, or boarding with a reptile-experienced vet or shop. Never a general boarding kennel
A hatchling or juvenile	Shift every row one tier stricter. Young tortoises are soaked more often and eat daily

TRANSPORT

A ventilated box lined with damp paper towel, no loose substrate, no water dish. Warm, 75 to 80°F (24 to 27°C), dark and quiet, belted in rather than loose on a seat. Never in the trunk or a parked car. On a move it travels with you, and its enclosure is running at the far end before it comes out.

THE FORTY-YEAR VERSION OF THIS PAGE

A tortoise outlives jobs, houses and relationships. Naming a caregiver is the first step, not the plan: they have to agree, know the animal, and be written into your estate papers with a second name behind them. Money does not travel with a pet, because the law treats an animal as property and property cannot inherit. A pet trust does, and all fifty states and the District of Columbia now have pet trust laws, so this is a question for an estate attorney. Leave a care file with the animal, and rehome through a rescue rather than a free listing. Write the name on page 28.

Pet-sitter sheet

Fill it in and leave it by the enclosure. Assume the sitter has never kept a reptile.

Tortoise name _____ Sex _____

Owner _____ Phone _____

Reptile vet _____ Phone _____

Emergency clinic _____ Phone _____

Backup contact _____ Phone _____

Normal weight (g) _____ Visit every _____

Each visit, please

- ☐ Empty, rinse and refill the water dish
- ☐ Scatter fresh greens on the stone, in a couple of places
- ☐ Remove yesterday's uneaten food and any droppings
- ☐ Read the cool-end thermometer (70s°F). If shown how, aim the infrared gun at the basking spot (95 to 100°F). They are meant to read differently
- ☐ Damp the moss in the humid hide if it feels dry
- ☐ Look at the tortoise walking. Text us a photo

PLEASE NEVER

Turn the lamps off, or unplug anything. The lights are on timers and the tortoise needs them.
 Move the tortoise outside, or take it out of the enclosure except as we have shown you.
 Let a dog or cat near it, even in the same room.
 Feed it fruit, meat, bread, dog or cat food, or anything from the never list below.
 Put it on its back, or lift it by a leg.

Never feed

Fruit, any meat or animal food, dog or cat food, bread, pasta, cereal, iceberg lettuce, avocado, rhubarb, or any plant from a garden. Only what is in the box in the fridge that we have labeled.

Call us if you see

- Breathing with its mouth open, or bubbles at the nose
- Not moving, or not coming out to bask at all
- A crack or bleeding anywhere on the shell
- Food untouched across two visits
- A lamp that has stopped working
- Anything that just seems different from last time

Where the food is _____

Which lamps are on timers, and when _____

Anything else _____

Owner log

Weigh weekly at the soak, or monthly once the animal is a settled adult, and always at the same point in the routine. Photocopy the page before you start; forty years is a lot of rows. Sudden weight loss in a tortoise is always significant, because these animals lose weight slowly by nature, and a chart is what turns "it looks a bit smaller" into a number a vet can use.

Equipment, supplement & vet log

The bulb date is the most important line on this page.

Equipment

ITEM	FITTED	REPLACE BY	NOTES
UVB tube (12 months)			
Basking bulb			
Substrate fully replaced			
Thermostat / timer checked			

Supplements & soaks

WEEK	CALCIUM (NO D3)	D3, IF VET ASKS	MULTIVITAMIN	SOAK (JUVENILES TWICE)
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐

Veterinary

DATE	REASON	WEIGHT (G)	FINDINGS, TREATMENT, FOLLOW-UP

Enrichment, ticked monthly

- ☐ Scatter feeding in new places
- ☐ Living forage to crop
- ☐ Decor rearranged, hides left alone
- ☐ Supervised outdoor time
- ☐ Dig area refreshed and dampened
- ☐ New terrain: stone, slope or log

Glossary

Terms and abbreviations used earlier in this book, in the sense they were used.

Brumation	The reptile equivalent of hibernation. A period of cold dormancy at 35 to 50°F, 10 to 14 weeks at most in this species, and optional indoors. Page 23
Carapace	The upper shell
CDC	Centers for Disease Control and Prevention, the US public health agency
CFR	Code of Federal Regulations, the official book of US federal rules. Page 25
CITES	Convention on International Trade in Endangered Species, the treaty that controls moving listed animals across borders. Page 26
Cloaca	The single opening for waste and reproduction. Cloacolith: a hardened urate blockage in it
CPW	Colorado Parks and Wildlife, the state agency whose rules ban this species in Colorado. Page 25
EAZWV	European Association of Zoo and Wildlife Veterinarians
FDA	Food and Drug Administration, the US agency behind the four-inch rule. Page 25
Gravid	Carrying eggs. Fertile or not
Herpesvirus	A viral disease of tortoises with no cure and high contagion. One published outbreak killed 75% of the group it reached. Page 19
Hexamita parva	A parasite of the kidney and urinary tract found frequently in this species. Page 21
IUCN	International Union for Conservation of Nature, which keeps the Red List of threatened species. Page 26
MBD	Metabolic bone disease, from a calcium and phosphorus imbalance, inadequate UVB, or both. Page 18
MSD	Merck Sharp & Dohme, the name the Merck Veterinary Manual uses outside the US and Canada
Mycoplasma	The bacterium behind chronic upper respiratory tract disease in tortoises: the runny nose that never fully clears. Page 19

REFERENCE

Glossary, continued

Terms and abbreviations, P to the end of the alphabet.

PCR	Polymerase chain reaction, a lab test that detects a virus's genetic material in a swab, so it can find herpesvirus in a tortoise that looks well. Page 21
Plastron	The lower shell
Pyramiding	Raised, cone-shaped scutes. Not completely understood; linked to temperature, humidity and calcium, phosphorus or protein imbalance during growth. Permanent. Page 18
SCL	Straight carapace length. The standard measurement, taken in a straight line rather than over the curve. Page 4
SCUD	Septicemic cutaneous ulcerative disease, an older term for severe shell infection. Page 20
Scute	One of the keratin plates covering the shell. They grow at their margins, which is why the shell is a record
T5 HO	T5 means a fluorescent tube five-eighths of an inch across; HO stands for high output. The slim, strong tube that is the right format for reptile UVB. Page 7
Units	in = inches, ft = feet, sq ft = square feet, cm = centimeters, mm = millimeters, g = grams, K = kelvin, °F/°C = degrees Fahrenheit/Celsius
Urates	The white, semi-solid crystalline waste a reptile passes instead of liquid urine. The best hydration gauge an owner has. Page 24
UV	Ultraviolet light, the invisible part of sunlight beyond violet. UVB is one band of it. Page 7
UVB	Ultraviolet B, the part of sunlight a tortoise needs on its skin to make vitamin D3 and absorb calcium. Glass filters out most of it and bulbs lose it with age. Page 7
UVI	Ultraviolet index, a number for how strong UV light is at one spot, read with a UV index meter. Target 3.0 to 4.0 at the basking site. Page 7
VCA	Veterinary Centers of America, a US chain of animal hospitals
21 CFR 1240.62	Title 21 of the Code of Federal Regulations, section 1240.62, the Food and Drug Administration's four-inch rule. It restricts commercial sale, not ownership. Page 25

REFERENCE

Sources

Where the numbers in this book came from.

Veterinary and specialist sources

SOURCE	WHAT IT SUPPLIED
VCA (Veterinary Centers of America) Animal Hospitals, <i>Common Diseases of Tortoises, Owning a Tortoise and Housing Box Turtles</i>	Metabolic bone disease, pyramiding, hypovitaminosis A, respiratory infection, abscesses, shell infections and SCUD, shell trauma, cloacoliths and parasites. Also the outdoor pen figures on page 9: the buried-barrier depths, the enclosed shelter, and the rule that water must never sit higher than the animal's head
Tortoise Trust, <i>Captive Care of the Russian Tortoise</i> and <i>Hydration and Your Tortoise</i>	Maximum verified size, sexual dimorphism, the burrow temperature figures, the substrate depth for a pallet, the 80-plant wild diet and the ninety-days-a-year feeding pattern, the low protein and no fruit rule, and the wild-caught trade
ReptiFiles, <i>Russian Tortoise Care Sheet</i>	The 7×3.5 ft adult enclosure minimum and per-additional-animal space, the UVB bulb, mounting distances and the UVI 3.0 to 4.0 target, the 40 to 75% juvenile humidity range, and the substrate depth
Merck Veterinary Manual (published as the MSD Veterinary Manual outside the US and Canada)	The preferred optimal temperature zone framing behind the cold-tolerance figures on pages 6 and 33, tortoise nutrition, and the external sexing cues on page 15
European Association of Zoo and Wildlife Veterinarians (EAZWV), <i>Herpesvirus Infection in Tortoises</i> , and Marenzoni et al., <i>PLOS ONE</i> , 2018	The six-month quarantine minimum and the testing protocol on page 21, and the documented outbreak behind the mortality figure on page 19
Royal Veterinary College, and the Texas Veterinary Medical Foundation	The whole brumation protocol on page 23: which species may brumate, the age floor, the fast, the temperature band and its ideal, the duration ceilings, weighing every one to two weeks and the weight-loss action point
Food and Drug Administration (FDA) and Centers for Disease Control and Prevention (CDC)	The four-inch rule and its business exception on page 25, and the <i>Salmonella</i> hygiene rules on page 10
Segura et al., <i>Scientific Reports</i> , 2020	The raven predation threshold for young tortoises on page 9
IUCN Red List and CITES	The Vulnerable listing and the trade status

Beastly Facts guides drawn on: Russian tortoise enclosure setup, health issues, handling, enrichment, cost and legal, plus the complete UVB lighting guide, the T5 versus compact comparison, the reptile emergency plan, and the cross-species chelonian herpesvirus and quarantine, tortoise brumation, tortoise soaking, tortoise sexing and egg binding, herbivorous reptile safe plants, outdoor reptile housing, reptile stool and urates, salmonella and hygiene, and long-lived pet succession guides.

REFERENCE

Where the sources disagree

Printed rather than averaged, so you can see the shape of the argument.

Open disagreements

SUBJECT	THE TWO POSITIONS, AND WHAT THIS BOOK PRINTS
Herpesvirus mortality	The Tortoise Trust puts mortality in an unquarantined group at near total. The one published outbreak investigation, Marenzoni et al. 2018, recorded 75% of twenty exposed tortoises dying, fifteen animals. Page 19 prints the documented 75% as one outbreak and names the higher figure beside it. Neither changes what you do
Enclosure size	General care guidance gives 4×2 ft as the indoor minimum and about 7×3.5 ft as better; ReptiFiles gives 7×3.5 ft as the adult minimum plus 10 sq ft per additional tortoise. Page 5 prints 4×2 ft as the floor and 7×3.5 ft as the target
Humidity	General care guidance gives 30 to 50% ambient; ReptiFiles gives 40 to 75% for juveniles, following what a burrow actually does, with adults needing only a humid hide. Page 8 prints both, because they agree on the shape: a dry ventilated surface over a genuinely humid microclimate
The <i>Hexamita</i> dose	The Tortoise Trust publishes a metronidazole figure. This book no longer prints it. Dosing depends on the individual animal's weight, kidney function and stage of infection, so page 21 names the drug and leaves the number to a vet

REFERENCE

Version history

What changed in each edition, and why.

VERSION	DATE	WHAT CHANGED
1.0	Aug 2026	First edition, 21 pages on the earlier layout
2.0	Sep 2026	Full rebuild on the current template, 37 pages. New pages for outdoor housing, weeds and grazing, sexing and body condition, eggs and egg binding, brumation, droppings and hydration, the legal position, and six further health pages. Enclosure size, humidity and diet all rewritten around specialist sources, with disagreements printed rather than averaged. Budget itemized so the lines sum
2.1	Sep 2026	Cross-checked against the September 2026 cross-species reptile articles. The brumation protocol on page 23 was corrected throughout: 1% a month is normal rather than the abort threshold, the band is 35 to 50°F with a 41°F ideal, the duration is a ceiling of 10 to 14 weeks by size, the fast is 1 to 3 weeks at 70 to 80°F, and the age floor rose to about four years. Quarantine went to a six-month minimum with testing rather than time alone. The male plastron is concave, not convex. Egg binding is separated from normal gravid behavior by whether she stays bright and alert, not by a two-day clock. The <i>Hexamita</i> dose was removed and the jelly-like urine sign added. Soaking went to two or three times a week. Sources, disagreements and version history split onto their own pages
2.2	Sep 2026	Legal edition, and the corrections are substantive. The at-a-glance page and the legal page both said "no state bans it; New Jersey requires a permit." Colorado bans it and four states require a permit. Colorado clears tortoises only if native to the tropical and subtropical region, which its rule fixes at 26.0 degrees north to 26.0 degrees south, and a Central Asian steppe animal fails that test. Massachusetts, New Mexico, Delaware, New York City, Vermont and Minnesota were added to the table, and the catch-all row is 41 rather than 43, since Montana and Rhode Island are already named above it.
2.3	Oct 2026	Accuracy and house rules edition. The enclosure target is now 7×3.5 ft (210×105 cm), about 24.5 square feet, with 4×2 ft kept as the floor. Soaking is once a week for adults and twice a week for juveniles, never unattended. Calcium with D3 only when a vet asks, and no fruit at all. West Virginia joins the permit states, so the legal counts now read 42 with no rule, 5 permit, 4 conditional and 1 ban. Pyramiding is described as not fully understood, high-oxalate greens left the rotation, pen walls rose to about 2 ft with an inward lip, and thermostat probe and lamp clearance rules were added. Every abbreviation is spelled out, page pointers were trimmed, and the glossary, the legal page and this history each gained a page, for 41 in all.

Beastly Facts publishes fact-checked care references for keepers who would rather read one accurate thing than ten contradictory ones. This is a husbandry reference, not veterinary advice, and the legal page is a summary rather than legal advice.

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